



# Presentation 2: Time-series Trends

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## Presentation #2 Goals:

# Called Number Analysis

**Business Relevance:** use the volume of incoming calls to each intake numbers to determine if we should increase the staffing for certain lines or merge multiple lines

**Method:** filtered 300+ unique called numbers and kept only 20+ numbers for which we have a clear understanding of their services

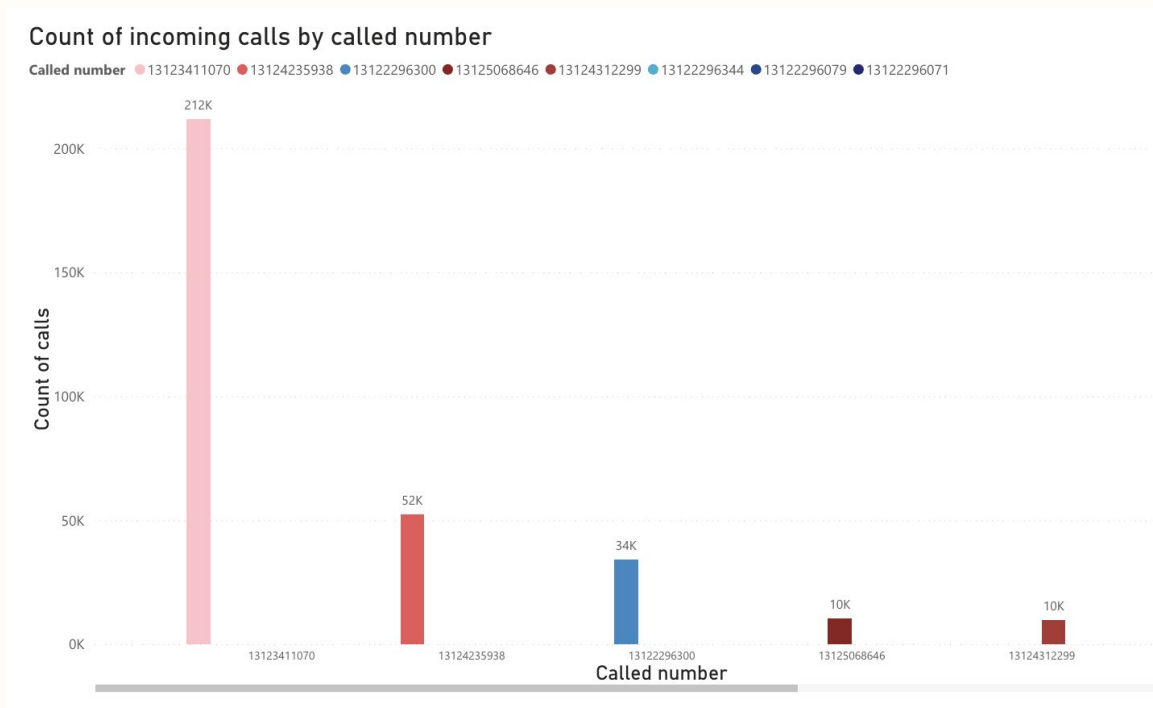
1. What is the distribution of incoming calls among the called numbers?
2. How does the number of incoming calls for each number vary over time?



# Volume of incoming calls by called number

## Top 5 numbers (#calls > 10k):

- Main Number
- Community Legal Clinics
- For those who already have an appointment → **internal routing number**
- Transfer to the English Main Menu → **internal routing number**
- Farmworker main number/ Migrant Legal Assistance Program

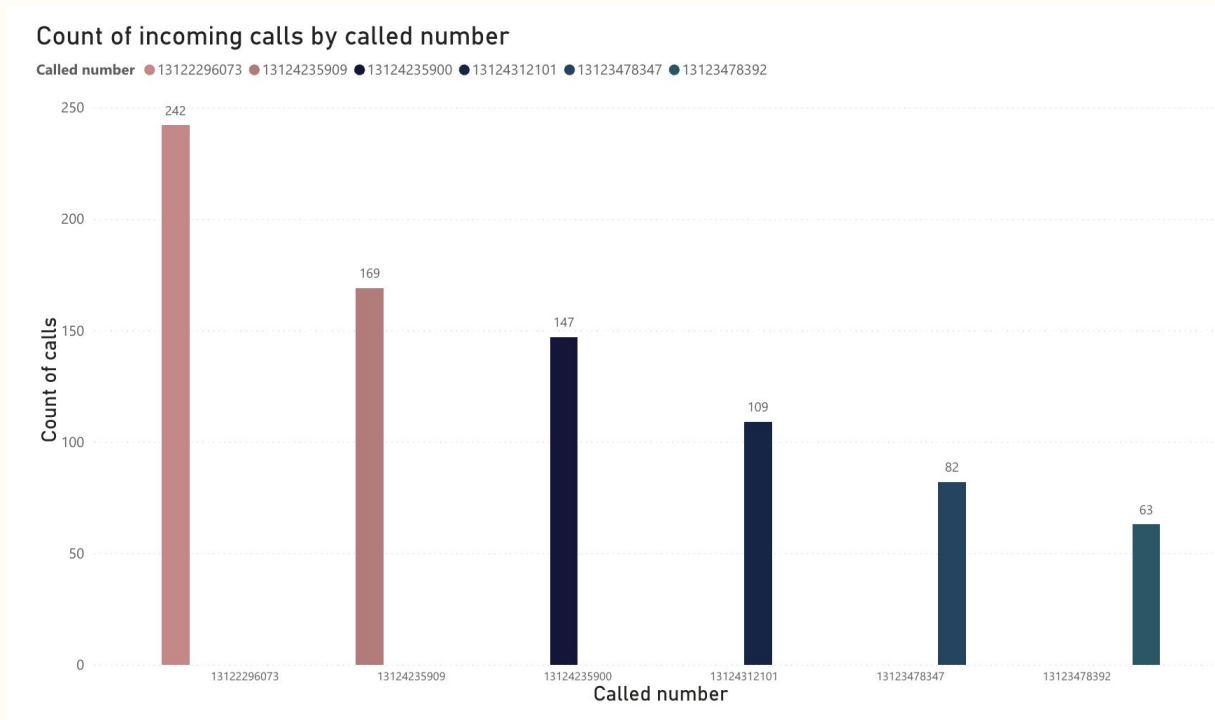


# Volume of incoming calls by called number

## Bottom 6 numbers

(#calls < 1000):

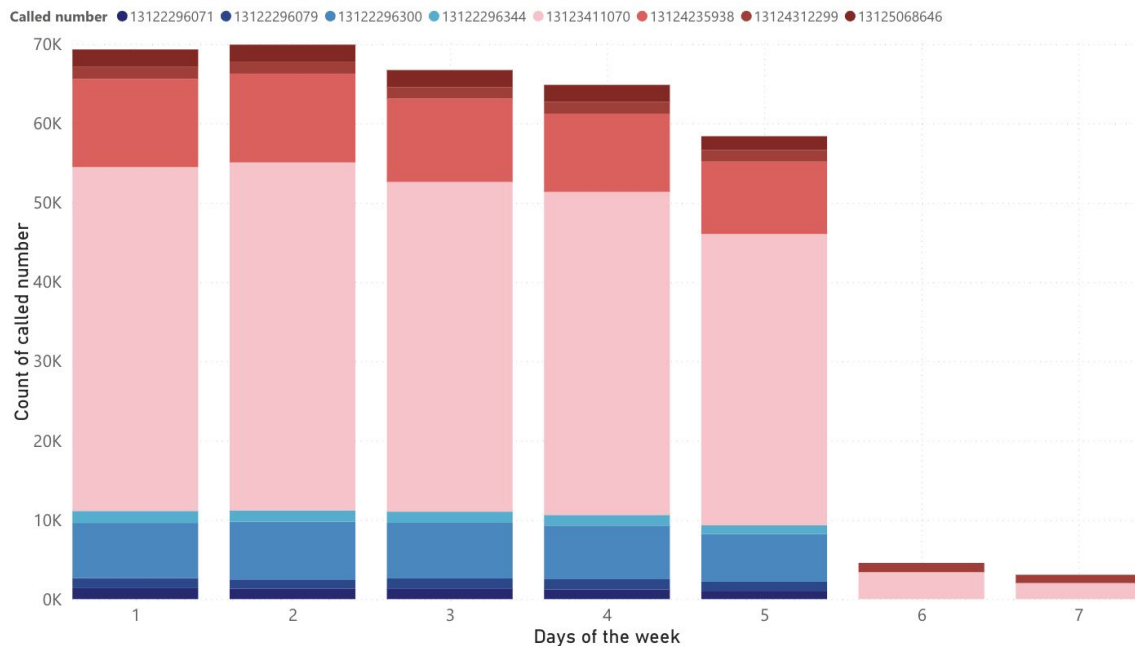
- Trafficking Survivors Assistance Program
- Fair Housing Intake VM
- CLASP (Free Legal Help for Sexual Violence Survivors)
- Order of Protection Appeals Project (Help for Domestic Violence Survivors)
- A2J Immigration (Illinois Access to Justice, immigrant education and legal services)
- Education Law Referrals VM



# Call volume by called number and days in a week

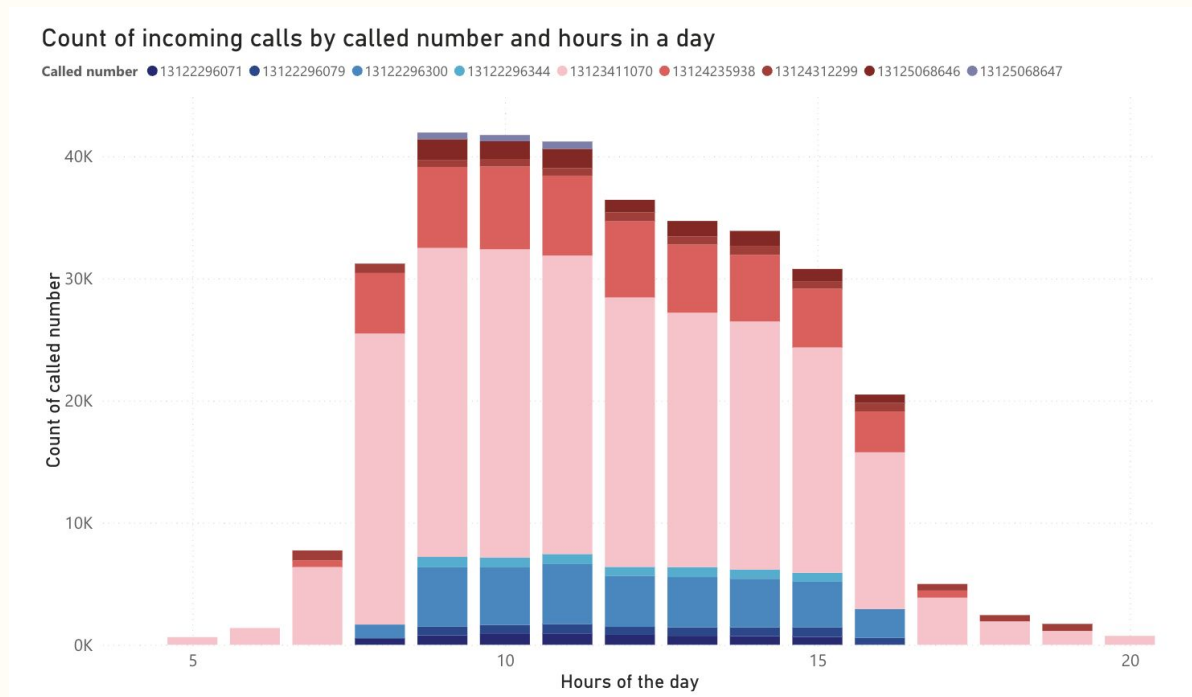
- Call volume reached a maximum on Monday and Tuesday, followed by a gradual decrease from Wednesday to Friday, and then a huge drop at weekends
- The top 3 numbers are still the Main number, number for Community Legal Clinics, and the internal routing number for transferring to English menus

Count of incoming calls by called number and days in a week



# Call volume by called number and hours in a day

- Call volume reached a maximum between 9AM and 12PM, followed by a gradual decrease and a huge drop between 4-5PM
- The proportion of incoming calls for each number stays stable throughout the day; the number of incoming calls for Main Number began to decrease from 12PM, but the number of calls for Community Legal Clinics is relatively the same from 9AM to 3PM



# Recommendations

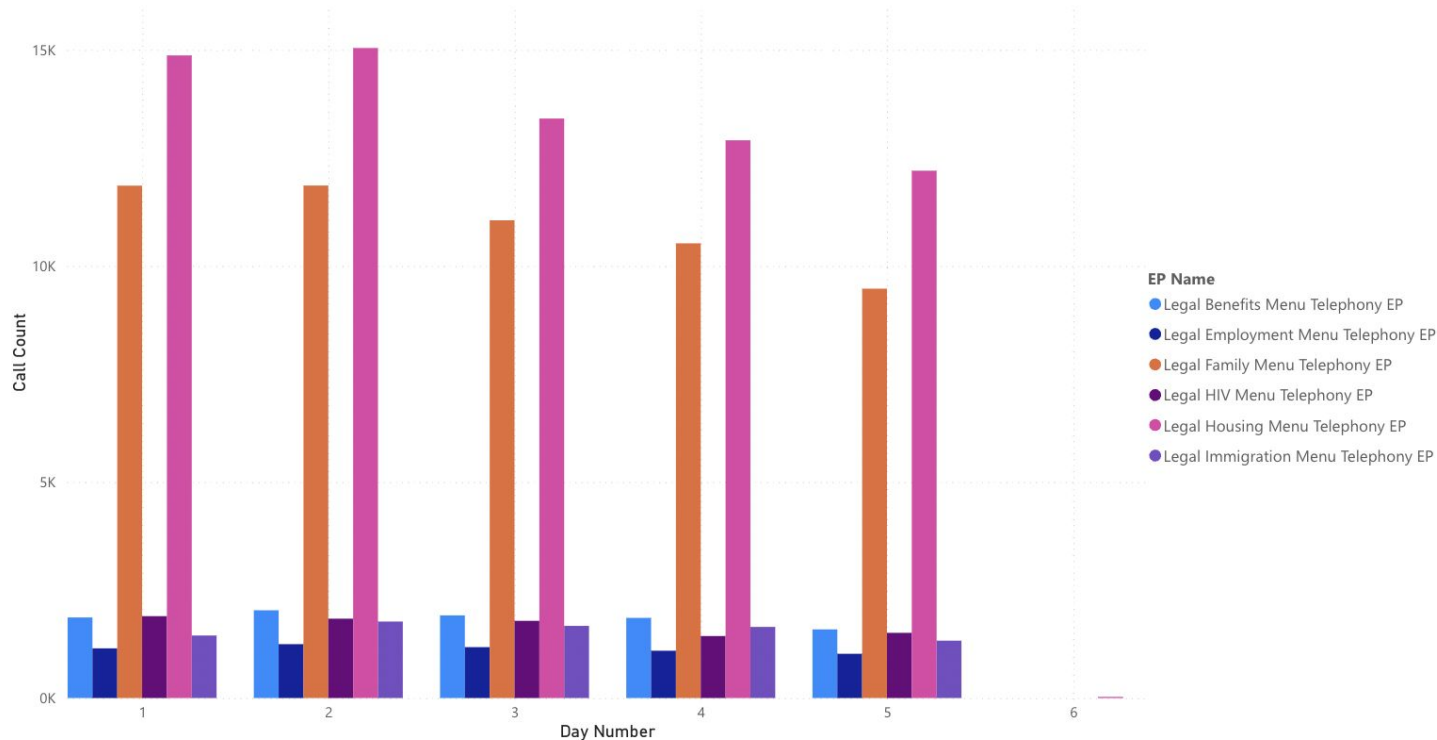
- Combine the intake lines for trafficking survivors, sexual violence survivors and domestic survivor into one line due to the low call volume of each line
- Combine the intake line for A2J Immigration with Farmworker main number/Migrant Legal Assistance Program

## Next Steps

- Map call flows and transfer patterns
  - Connect internal routing numbers to their corresponding intake lines to visualize the complete workflow of incoming calls
  - Analyze how calls are transferred between lines (using “Correlation ID”) to identify common transfer paths, entry points, and opportunities to streamline the routing process
- Clarify outbound call structure
  - Investigate how outbound calls are logged, given the absence of a clear “calling number” column.

# Frequency of Calls by Day of Week and Hours

Call Count by Day Number and EP Name



## Findings:

- Calls peak on Monday and Tuesday, gradually decline through rest of the week

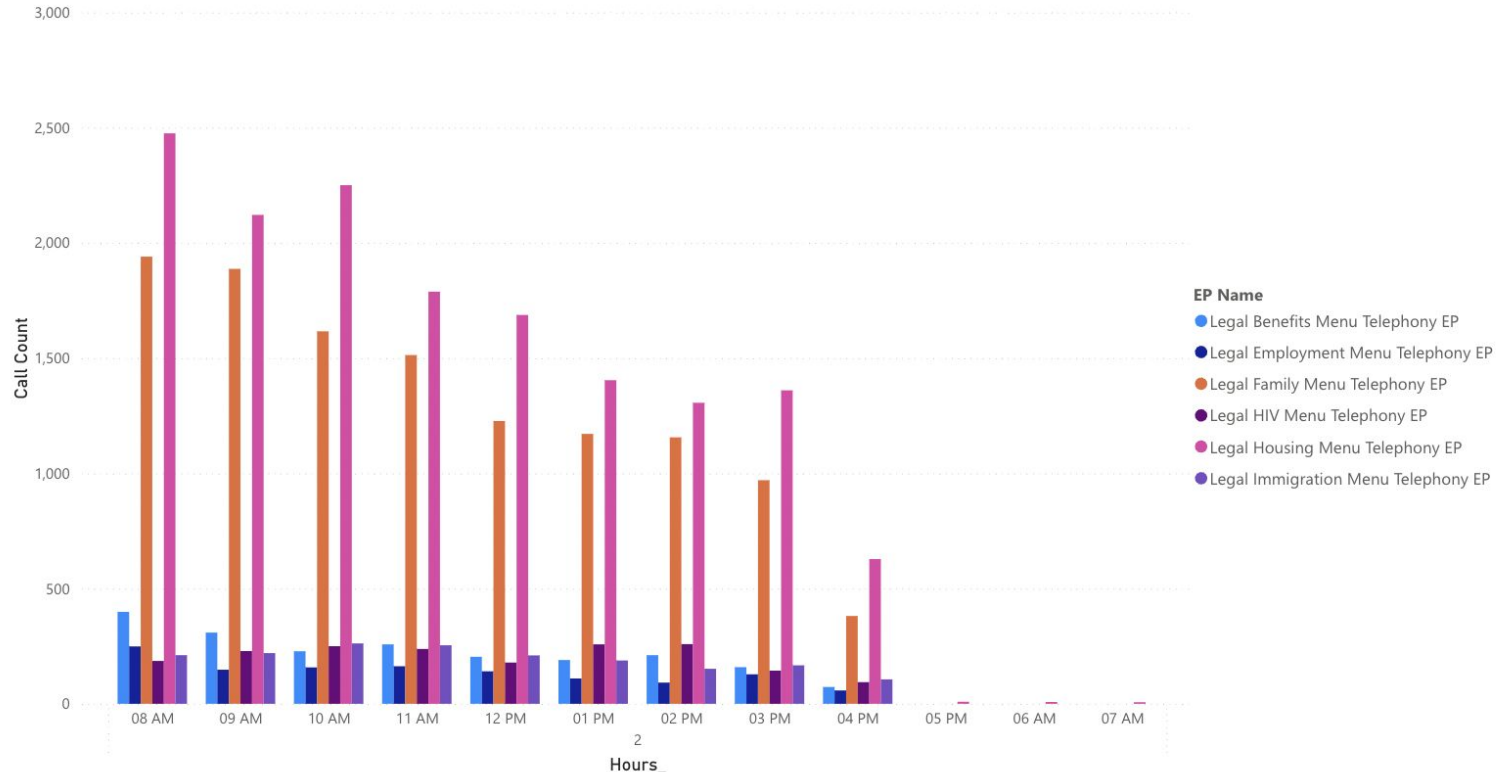
## Business Relevance:

- Staff to full capacity on M/T, have flexibility for other days to cut costs
- Automated message telling customers to call back on a less busy day



# Frequency of Calls by Day of Week and Hours (Drill-Down)

Call Count by Day Number, Hours\_ and EP Name



## Findings:

- Calls are highest in the morning hours, decline throughout rest of the day

## Business Relevance:

- Staff to full capacity in the morning, have flexibility after to cut costs
- Leave an automated message of best time of day to call back

# Recommendations & Next Steps:

## **Recommendations:**

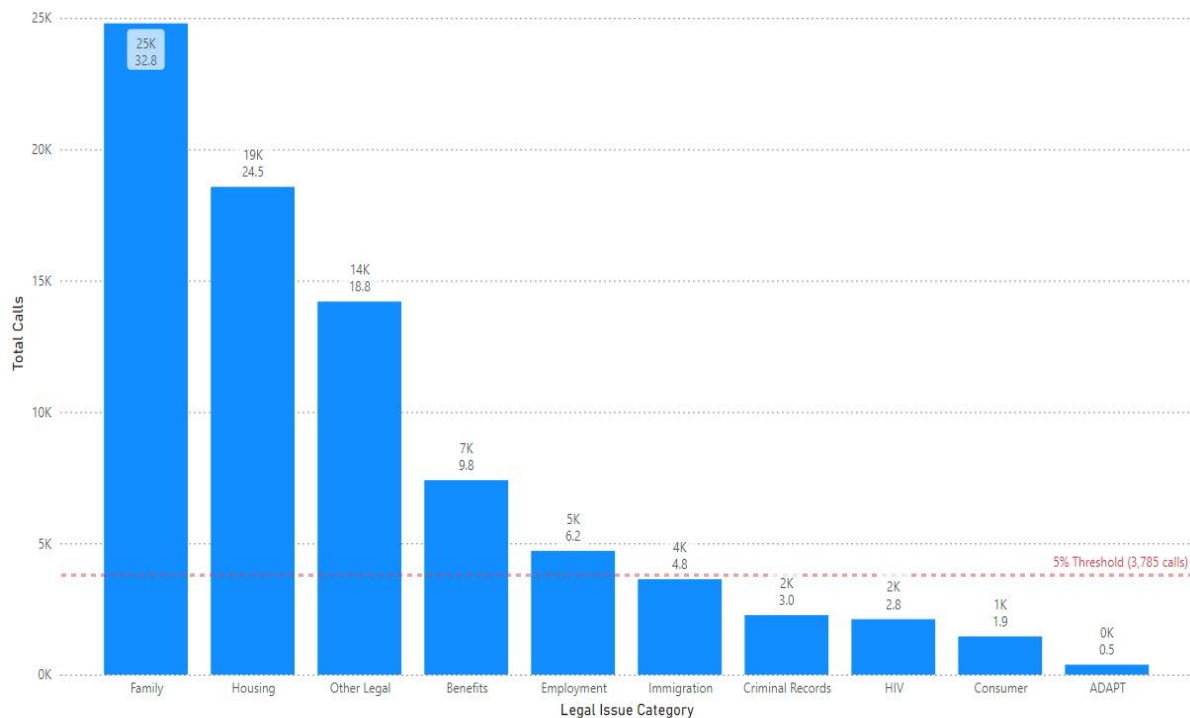
- Staff at full capacity on Mondays and Tuesdays from 8am to 10am for the family and housing lines
- More flexibility at other times to cut costs

## **Further Investigation:**

- Detailed analysis on the quality of calls across day of the week and hours of the day through call outcome column for a better understanding of call overflow
- Employment line is the lowest compared to all the other lines– why is this?

# Call Frequency for Legal Menu Categories

Legal Issues Categories: Call Frequency with 5% Consolidation Line



## Business Relevance:

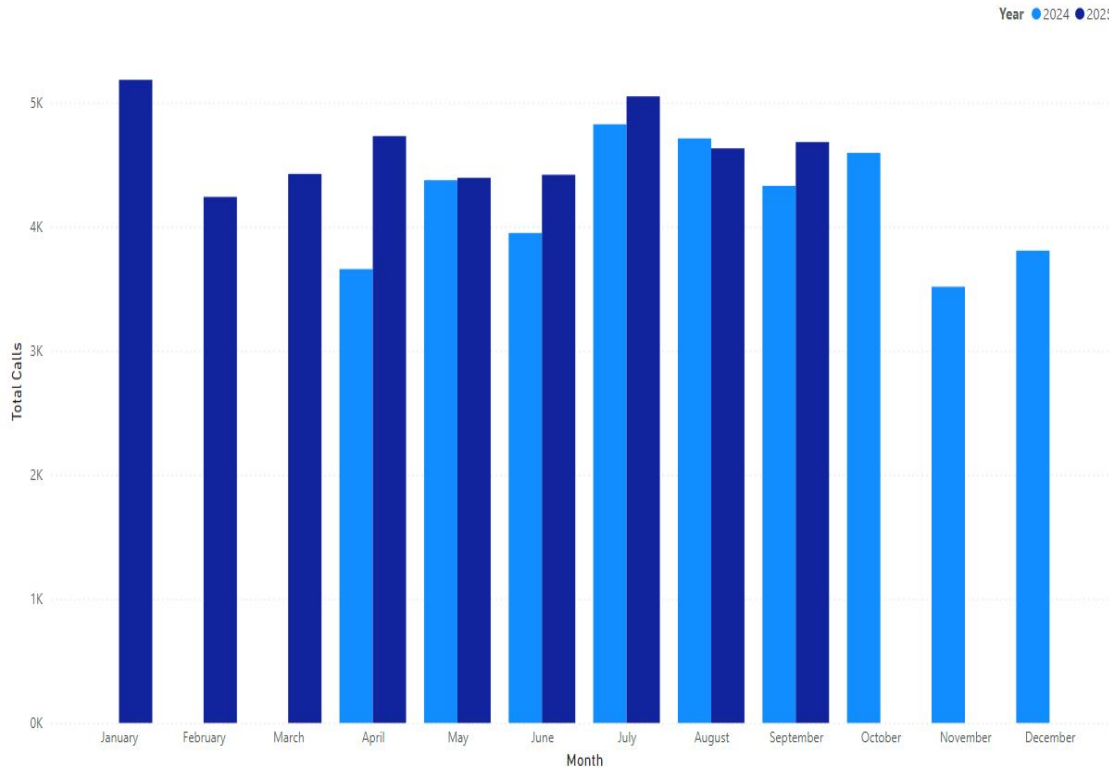
- This graph identifies which legal issue categories receive the highest call volumes and which fall below the 5% threshold.

## Findings:

- Family, Housing, and Other Legal make up the majority of calls
- Immigration, Criminal Records, HIV, Consumer, and ADAPT fall below 5% of total calls

# Legal Issues Call Volume per Year

Monthly Call Volume Comparison per Year (Legal Issues)



## Business Relevance:

- This graph tracks monthly call volume trends across fiscal years (October 2024 – September 2025), in order to identify seasonal/monthly trends.

## Findings:

- There tends to be consistent increase in calls per each month in 2025, but not too much of a jump, except for April
  - May correlate with tax season
- January had the highest call volume

# Recommendations & Next Steps:

## **Recommendations:**

- Combine the six legal issue categories that are called the least (Employment, Immigration, Criminal Records, HIV, Consumer, and ADAPT) into a single menu option.
- Prioritize staffing during January and April.

## **Further Investigation:**

- Create a drill-down of call patterns for month/year within sub-menus for Family, Housing, and Other Legal menu.

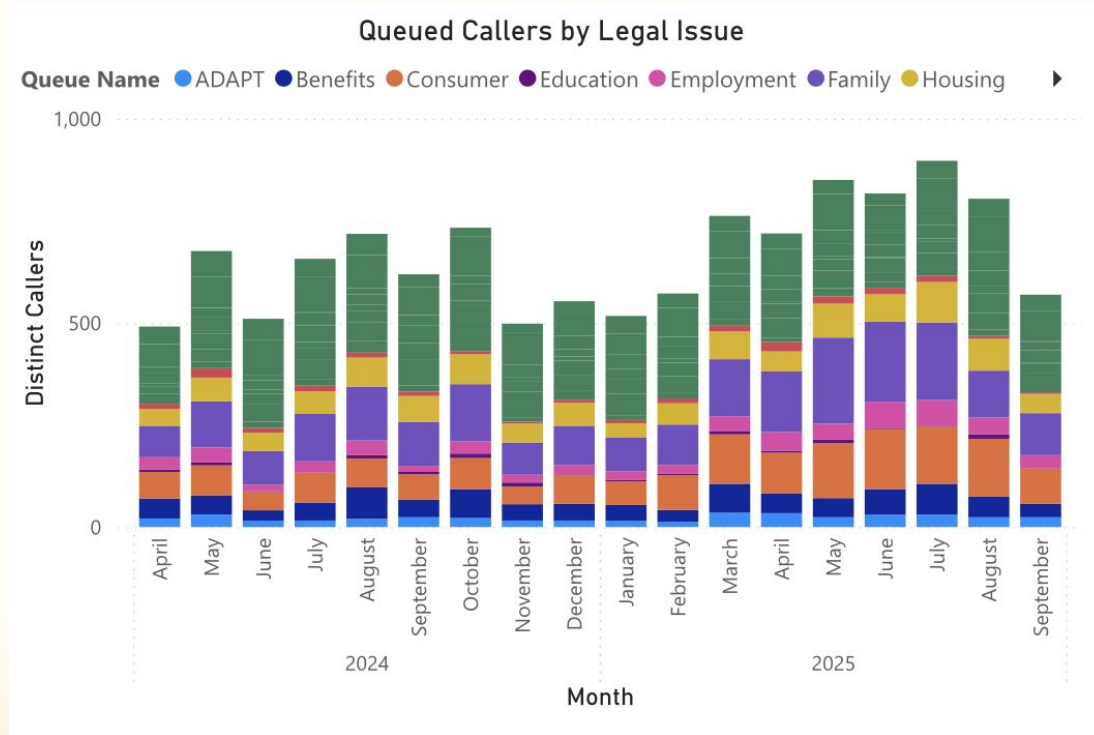
# Queue Analysis

## Presentation #2 Goals:

1. Update Queue Demography Dashboard based on instructor feedback
2. Open vs. closed queue analysis during business hours
3. Queue wait time analysis



# Updated Queue Demography Bar Chart



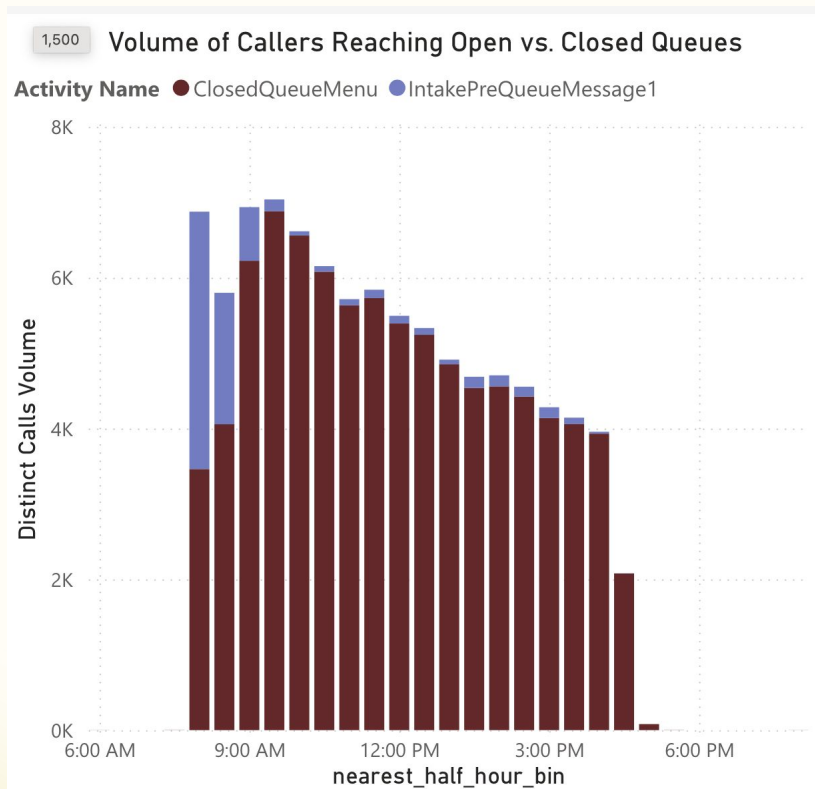
## Key Changes

- Drill Down functionality
- Included subseniors (previously excluded)

## Key Findings

- Subseniors constitute significant proportion of queued callers

# Open vs. closed queue analysis during business hours

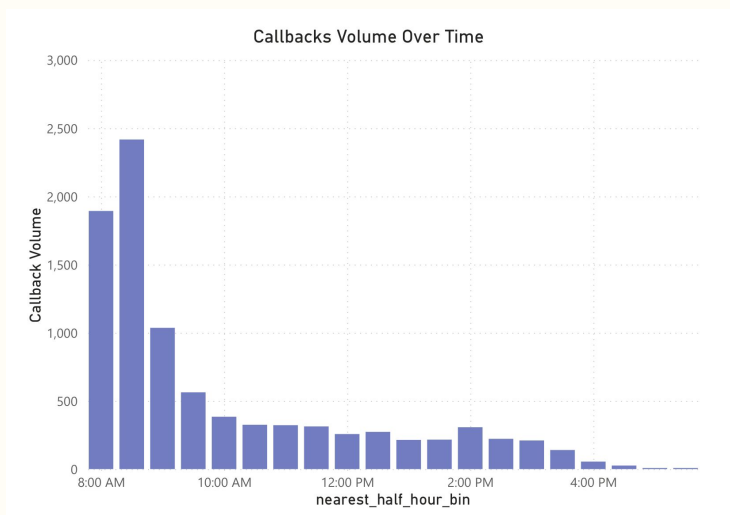


## Key Findings

- Queue tends to only consistently open from 8–8:30 am, sometimes 8:30–9 am
  - Sporadic openings later in the day
- High volume of callers reaching closed queue later in the day

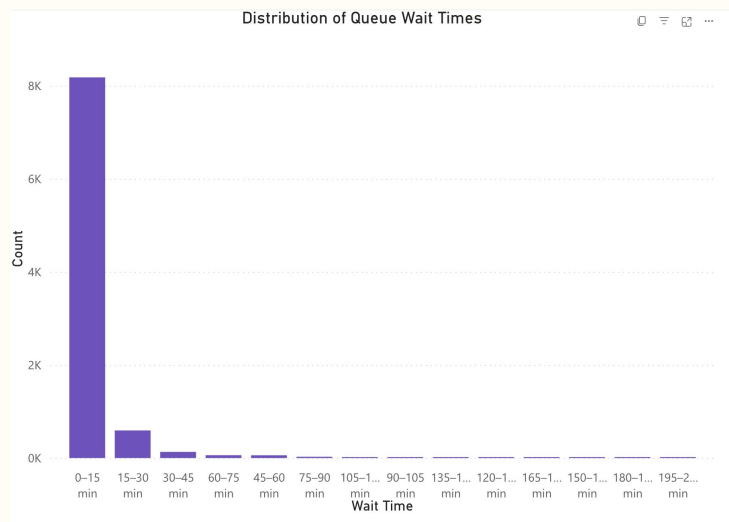


# Queue Callback and Wait Times



## Key Findings

- Majority of callbacks for callers who stayed on the call occurred between 8–9 am, lesser volume throughout the day



## Key Findings

- Vast majority of callers who stayed on the call waited less than 15 minutes to talk to an intake specialist.

## Next Steps

- If hang up/callback is a different contact session ID, will try to analyze the success rates of these callbacks.
  - Low success rate, this may indicate that callers feel discouraged by long queue wait times.
- Analyze the paths that people who reach an open queue take vs. those who end up reaching a closed queue.
  - Seniors may be more likely to have time flexibility to reach an open queue whereas those who have a strict work M-F schedule/other responsibilities are unable to catch the open queue.

# Questions for Cynthia

- How does the clinic triage seniors compared to non-seniors?
- What does a callback retry mean?
- Queue wait times are by contact session ID— if the caller leaves their number for a call back, will their contact session ID be different?